

Many paths lead to the summit

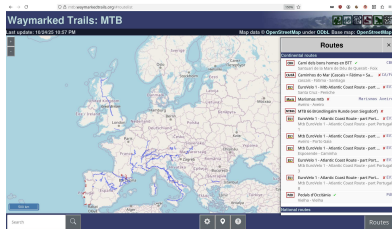
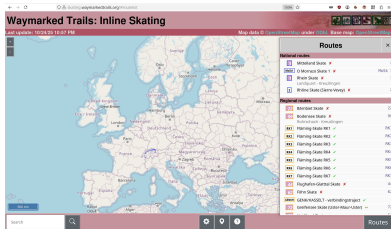
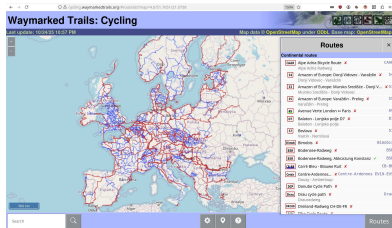
Complex hiking and cycling routes in OpenStreetMap

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15th Nov 2025

State of the Map Europe 2025
Dundee

Hiking, Cycling, MTB, Skating Routes in Europe



A Simple Hiking Route

Tags

Key	Presence	Value	Explanation
<code>type</code>	 Required	<code>route</code>	Indicates this relation represents a route.
<code>route</code>	 Required	<code>road</code> / <code>inline skates</code> / <code>evacuation</code> / <code>detour</code> / <code>bicycle</code> / <code>mtb</code> (mountainbike) / <code>foot</code> / <code>hiking</code> / <code>running</code> / <code>bus</code> / <code>trolleybus</code> / <code>waterway</code> / <code>ferry</code> / <code>canoe</code> / <code>train</code> / <code>subway</code> / <code>light rail</code> / <code>railway</code> / <code>tracks</code> / <code>tram</code> / <code>funicular</code> / <code>horse</code> / <code>ski</code> / <code>snowmobile</code> / <code>piste</code>	A road (e.g., the ways making up the A7 auto route), bicycle route, hiking route or whatever route (see also #Route types (route) and #Others).
<code>name</code>	 Important	A name	The route is known by this name (e.g., <code>Jubilee Cycle Route</code> , <code>Pembrokeshire Coastal Path</code>).
<code>ref</code>	 Important	A reference	The reference by which this route is known; e.g., <code>A7</code> , <code>NCN 11</code> , <code>Citi 4</code> . Recommended if no parent relation <code>route master=*</code> exists. Otherwise, it is optional.
<code>network</code>	 Important	<code>ncn</code> / <code>rcn</code> / <code>lcn</code> / <code>nwn</code> / <code>rwn</code> / <code>lwn</code> / ...	A wider network of routes of which this is one example. For example, the UK's national cycle network or a local cycle network.
<code>operator</code>	 Optional	Operator name	The route is operated by this authority, company, etc; e.g., <code>Stagecoach Cambridge</code> , <code>Eurostar</code> .
<code>state</code>	 Optional	<code>proposed</code> / <code>alternate</code> / <code>temporary</code> / <code>connection</code>	Sometimes routes may not be permanent (i.e., diversions), or may be in a proposed state (e.g., UK NCN routes are sometimes not official routes pending some negotiation or development). Connection is used for routes linking two different routes or linking a route with for example a village centre.
<code>symbol</code>	 Optional	Symbol description	Describes the symbol that is used to mark the way along the route; e.g., <code>Red cross on white ground</code> for the <code>Frankenweg</code> in Franconia, Germany.
<code>colour</code>	 Optional	A color name or a <i>hex triplet</i>	Colour code noted in <i>hex triplet</i> format. Especially useful for public transport routes. Example: <code>#008080</code> for teal colour.
<code>description</code>	 Optional	A short description	What is special about this route.
<code>distance</code>	 Optional	Distance	The nominal distance covered by the route. For users' information and automatic evaluation; e.g., of completeness. The default unit is km. Use dot as decimal separator. If a unit is specified, it should be separated from the number by a space and use a value as described on the Units page.
<code>ascent</code>	 Optional	Ascent	The ascent covered by a route (default units are metres; specify others explicitly). If a route has start and end point at different altitude use descent too.
<code>descent</code>	 Optional	Descent	The descent covered by a route (default units are metres; specify others explicitly). Use it only if it differs from the ascent (different altitude at start/endpoint of a route).
<code>roundtrip</code>	 Optional	<code>yes</code> / <code>no</code>	Whether or not the route start and ends at the same place; i.e., after traversing the route once, one has returned to the start.

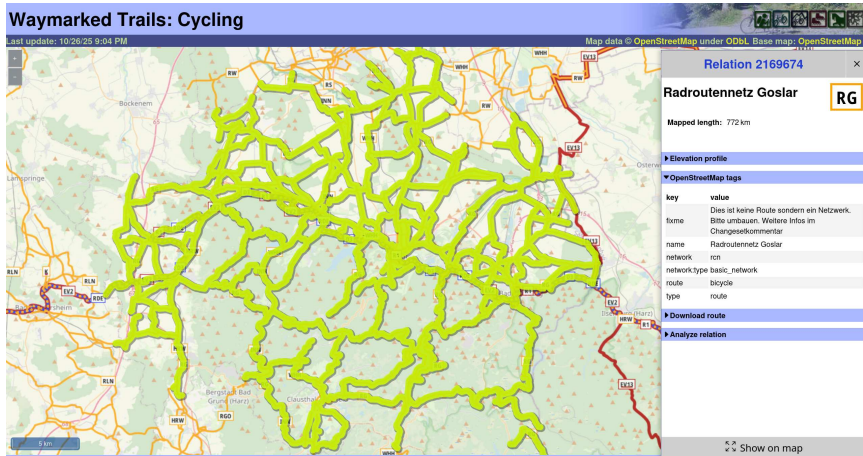
Content warning

Slides may contain strong opinions

... on what's correct tagging

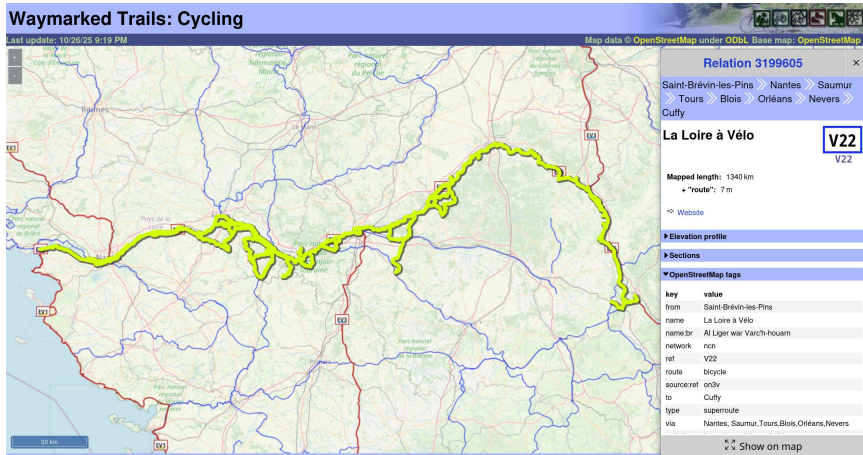
To Sort or not to Sort

The Illusion of Continuity



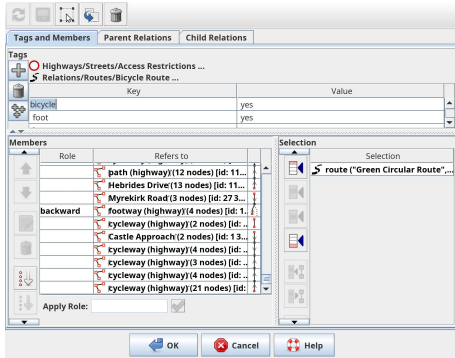
Not all route relations are linear.

The Illusion of Continuity



Route relations have a inherent direction.

Tool Support For Finding Errors



Waymarked Trails: Cycling

Last update: 10/26/25 9:19 PM

The screenshot shows a map of Dundee, Scotland, with a blue line indicating a cycling route. The route starts at the top left, goes through the city center, and ends at the bottom right. The map includes labels for various locations such as Burnside of Duntrune, Ballumbie Castle Estate, Ballumbie, Baldovie, Douglas, Mid Craigie, Linlathen, Miln's Mains, Fairmuir, Kirkton, Downfield, The Glens, Lawside, Hilltown, Stobswell, Craiggiebank, Dudhope, Blackness, Dundee, West End, Newport-on-Tay, and Westhaven. The map also shows a scale bar for 2 km.

Routes

Continental routes

Route	Status	Distance
EV12 EuroVelo 12 - North Sea Cycle Route	✗	EV12

National routes

Route	Status	Distance
77 NCN National Route 77	✗	77
777 NCN National Route 777	✗	777
1 Dover - Tain	✗	1

Regional routes

Route	Status	Distance
KCR Kingdom Cycle Route	✗	Inverkeithing - Forth Road Bridge

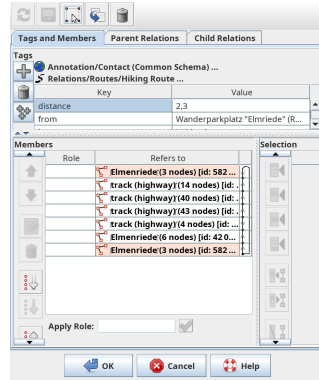
Other routes

Route	Status	Distance
GCR Green Circular Route	✗	
PCT Pine Cone Trail	↔	
RTWCR River Tay Way Cycle Route	✓	Kenmore - Strathtay - Dalguise - Dunkeld - Birnam - Bankfoot - Almondbank - Perth - St Madoes - Errol - Invergowrie - Dundee

Map data © OpenStreetMap under ODbL. Base map: OpenStreetMap

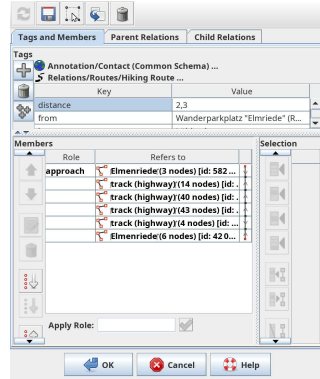
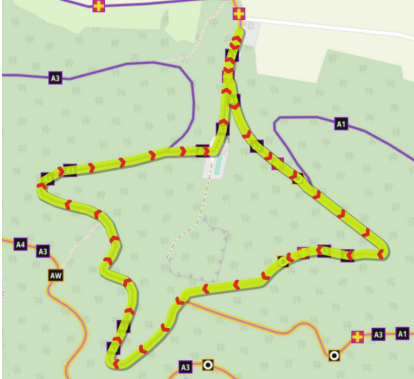
Balloon Routes

Roundtrips with Approaches



put balloon strings twice: at the beginning and, in inverse order, at end

Roundtrips with Approaches



map balloon strings once as an approach

Directional Routes

`oneway=yes`

Route can only be used in one direction.

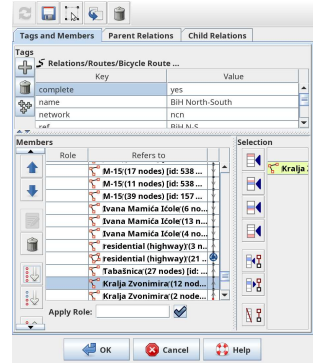
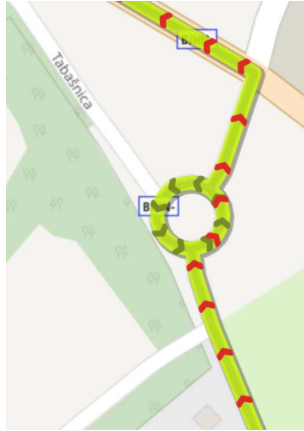
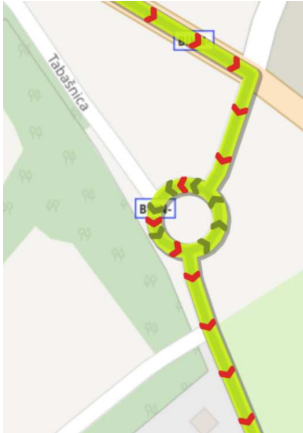
`signed_direction=yes`

Route is signed only in one direction.

Direction as per relation order.

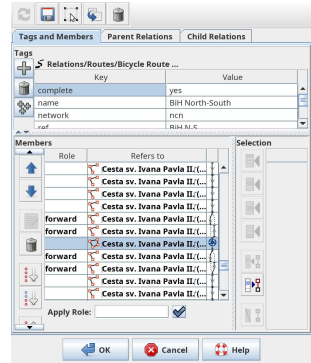
Roundabouts

The Simple Roundabout

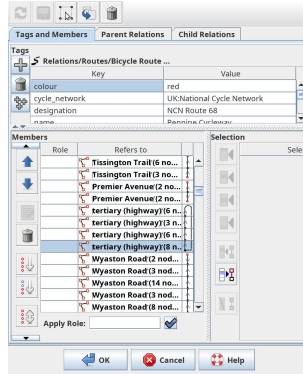


Leave them as one circular way tagged with `junction=roundabout`

Roundabouts with Split Approaches



Split Roundabouts



add ways actually traveled with forward/backward directly after each other

Navigating oneway roads

Forward and Backward

Waymarked Trails: Cycling

Last update: 10/29/25 10:30 PM

Map data © OpenStreetMap under ODbL. Base map: OpenStreetMap



Relation 20638 ×

Fleurier > Zürich

L'Areuse-Emme-Sihl 94
94

Fleurier - Zürich

Mapped length: 256 km

Official length: 260 km

Operator: Veloland Schweiz

↔ [Website](#)

► Elevation profile

► Sections

► OpenStreetMap tags

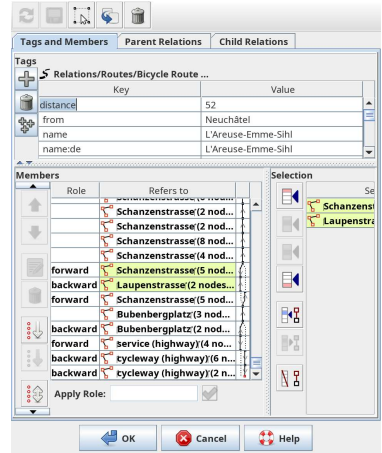
► Download route

► Analyze relation

↗ Show on map

Forward and Backward

- ▶ always comprises two subroutes starting and ending at the same point
- ▶ add one subroute first, then other *both in order of the relation*
- ▶ use forward/backward to describe which direction of the way to use



The Fork in the Route

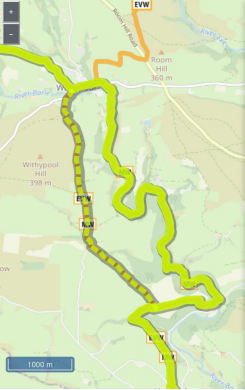
Alternatives



Alternatives

Waymarked Trails: Hiking

Last update: 10/30/25 9:15 PM



The map shows a green hiking route (Two Moors Way) winding through a landscape with hills and rivers. Key locations marked include Room Hill (360 m) and Withypool Hill (398 m). The route is highlighted in green with yellow markers.

Relation 228450

Two Moors Way

Mapped length: 150 km
+ "alternative": 11 km
Official length: 166 km

[Website](#) [Wikipedia](#)

Elevation profile

OpenStreetMap tags

key	value
length	166km
name	Two Moors Way
network	rwn
ref	MW
route	hiking
symbol	green M on top of black W
type	route
url	http://www.twomoorsway.org/

Map data © OpenStreetMap under ODbL. Base map: Op

[Show on map](#)

Tags and Members

Parent Relations Child Relations

Tags

Key	Value
length	166km
name	Two Moors Way
network	rwn
ref	MW
route	hiking
symbol	green M on top of black W
type	route

Members

Role	Refers to
	unclassified (highway)(13 nodes) ...
	footway (highway)(23 nodes) [id: ...]
alternative	bridleway (highway)(32 nodes) [id: ...]
alternative	Worth Lane(2 nodes) [id: 413 189...
alternative	Worth Lane(2 nodes) [id: 500 728...
alternative	Worth Lane(26 nodes) [id: 282 63...
alternative	Worth Lane(35 nodes) [id: 145 85...
alternative	Worth Lane(7 nodes) [id: 508 459...
alternative	Room Hill Road(4 nodes) [id: 121 ...]
alternative	New Bridge(2 nodes) [id: 121 90...
alternative	Room Hill Road(5 nodes) [id: 358 ...]
alternative	Room Hill Road(4 nodes) [id: 278 ...]
	tertiary (highway)(3 nodes) [id: 1...

Apply Role: ☐

Selection

Selection

Worth Lane(35 nodes)

OK Cancel Help

Branches

Waymarked Trails: Hiking

Last update: 10/30/25 9:15 PM

Map data © OpenStreetMap under ODbL. Base map: OpenStreetMap



Relation 6026933

Via Turonensis

655
GR 655

Mapped length: 1108 km

+ "branch": 716 km

+ "approach": 3.6 km

Operator: FFRP

Note: Cette relation contient la version de la Voie de Tours gérée par la FFRP, qui démarre depuis la Belgique. La version gérée par des associations jacquaires fait l'objet d'une autre relation.

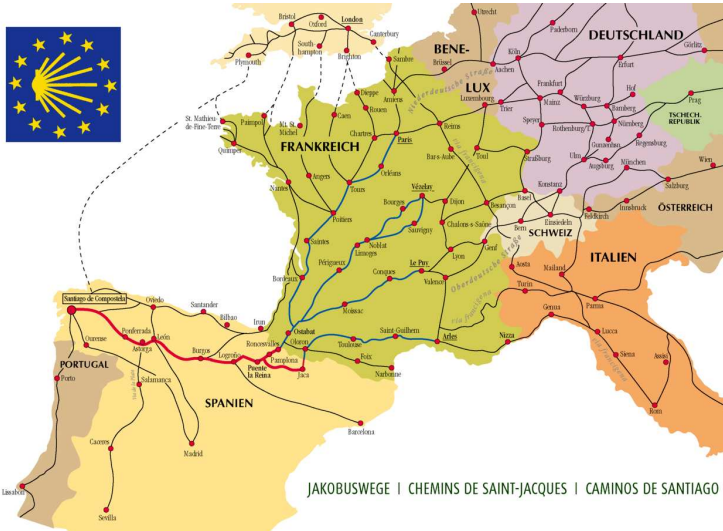
► Elevation profile

▼ Sections

-  Vieux-Reng - Maubeuge ✓
-  Maubeuge - Berlaymont ✓
-  Berlaymont - Landrecies ✓
-  Landrecies - Bohain ✓
-  Bohain - Saint-Quentin ✓

 Show on map

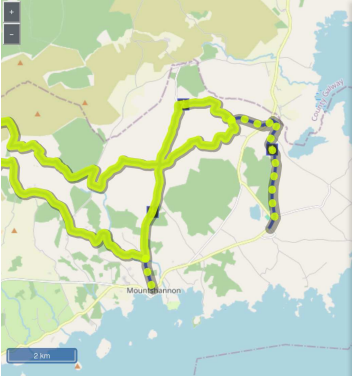
Branches



Appendixes

Waymarked Trails: Hiking

Last update: 11/2/25 4:41 PM



The map shows a yellow highlighted hiking trail in a rural area with green fields and some buildings. A scale bar at the bottom left indicates 2 km. The trail starts near a body of water and winds through the landscape.

Relation 14591638

Killaloe >> Killaloe

East Clare Way

Mapped length: 156 km
+ "approach": 13 km
Official length: 180 km
Operator: Clare Walks Ltd

[Website](#)

Elevation profile

OpenStreetMap tags

key	value
distance	180 km
from	Killaloe
name	East Clare Way
name:en	East Clare Way
network	nwn
operator	Clare Walks Ltd

[Show on map](#)

Map data © OpenStreetMap under ODbL. Base map: OpenStreetMap

Tags and Members

Annotation/Contact (Common Schema) ...
Relations/Routes/Hiking Route ...

Key	Value
distance	180 km
from	Killaloe
name	East Clare Way
name:en	East Clare Way

Members

Role	Refers to
approach	Kevin Parade (7 nodes) (id: 1 ...)
approach	L3004 (47 nodes) (id: 192 23...
approach	unclassified (highway) (5 no...
approach	unclassified (highway) (18 no...
approach	tertiary (highway) (71 nodes)
approach	Main Street (14 nodes) (id: 1 ...)
approach	track (highway) (3 nodes) (id: ...)
approach	track (highway) (4 nodes) (id: ...)
approach	path (highway) (6 nodes) (id: ...)
approach	track (highway) (5 nodes) (id: ...)
approach	R352 (8 nodes) (id: 1 099 689 ...)
approach	unclassified (highway) (43 no...
approach	unclassified (highway) (2 no...
approach	L4044 (5 nodes) (id: 1 099 68...
approach	Main Street (11 nodes) (id: 4...

Apply Role: ☐

Selection

Selection

route ("East Clare Way", 27...

OK Cancel Help

Appendixes

Waymarked Trails: Hiking

Last update: 11/2/25 4:41 PM

Map data © OpenStreetMap under ODbL. Base map: OpenStreetMap



Relation 72868

Eppingen >> Pforzheim

Eppinger-Linien-Weg  ELW

Eppingen - Sternenfels - Mühlacker - Pforzheim

Mapped length: 70 km
+ "alternative": 1.9 km
+ "excursion": 131 m

Official length: 75 km

Operator: Naturpark Stromberg-Heuchelberg

Note: Wanderweg wurde Ende September 2014 teilw. neu gestaltet sowie mit Skulpturen versehen. Letzte "alte" Relation vor der Änderung ist Version 209.

► Elevation profile

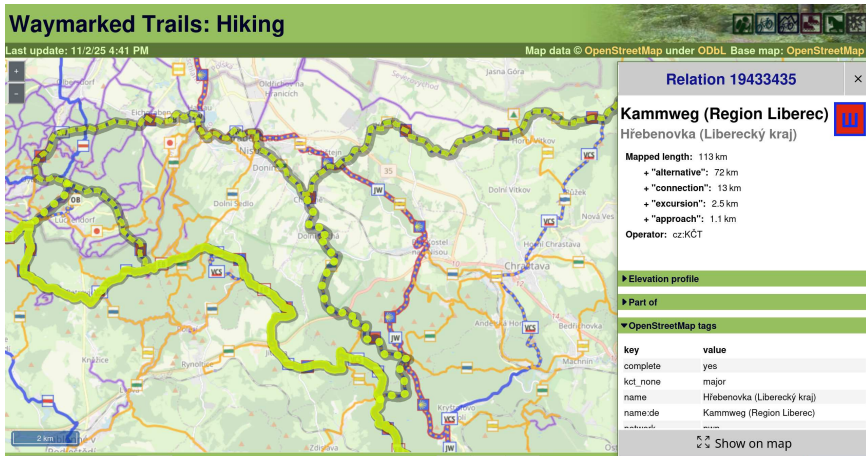
▼ OpenStreetMap tags

key	value
description	Eppingen - Sternenfels - Mühlacker - Pforzheim

Show on map

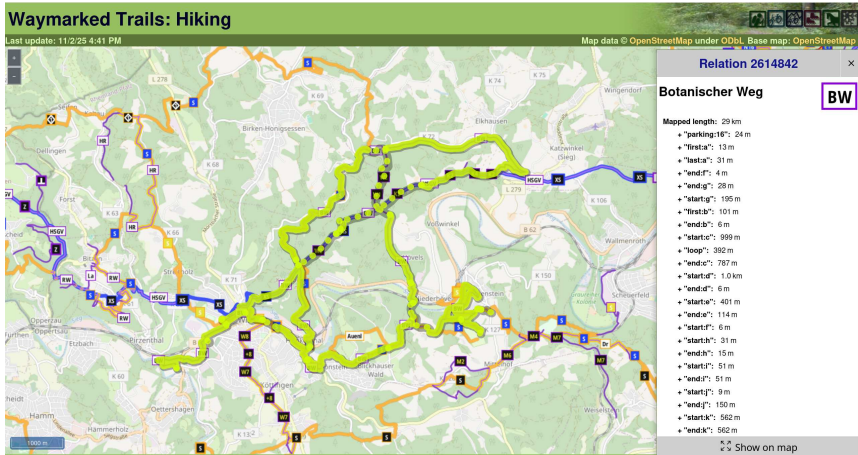
role = excursion

Appendixes



role = connection

Appendixes



role = ?

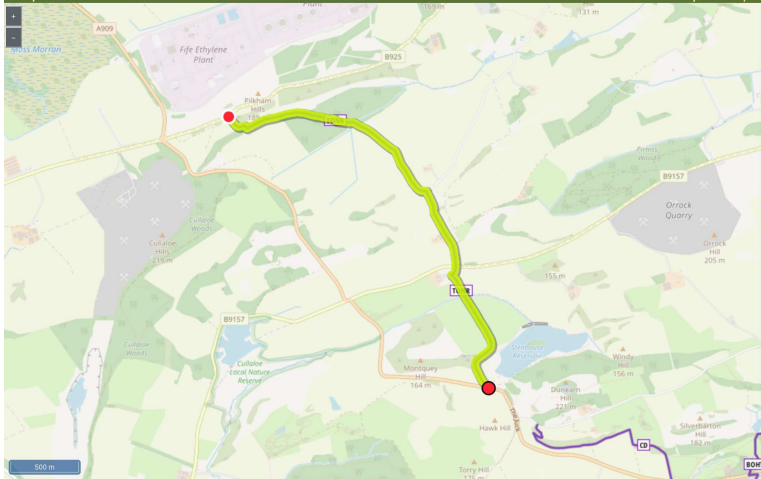
Relation Gardening

Analyse your relation

Waymarked Trails: Hiking

Last update: 11/2/25 2:35 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap



Relation 11671079

Website

Elevation profile

OpenStreetMap tags

key	value
name	The Old North Road
network	lwn
route	hiking
type	route
website	http://www.heritagepaths.co.uk/pathdetails.php?path=23

Download route

Analyze relation

This relation has 1 segments.

- Open end in the route.
- More than two segments of the route meet in this point.
- Point where route is interrupted because relation is not sorted. The red dashed lines shows where the route continues.

This shows the route segments as they were computed by waymarkedtrails from the relation, which may not always be correct. For more information how waymarkedtrails sorts routes, visit the help pages.

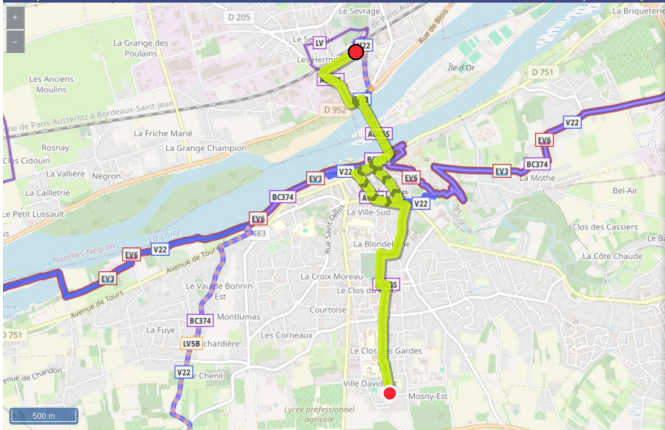
Show on map

Analyse your relation

Waymarked Trails: Cycling

Last update: 11/2/25 2:35 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap



Relation 17855203

construction	yes
name	Axe Gare-Cité Scolaire du SDC Amboise
network	lcn
route	bicycle
signed_direction	no
type	route

Download route

Analyze relation

This relation has 1 segments.

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Show on map

Search



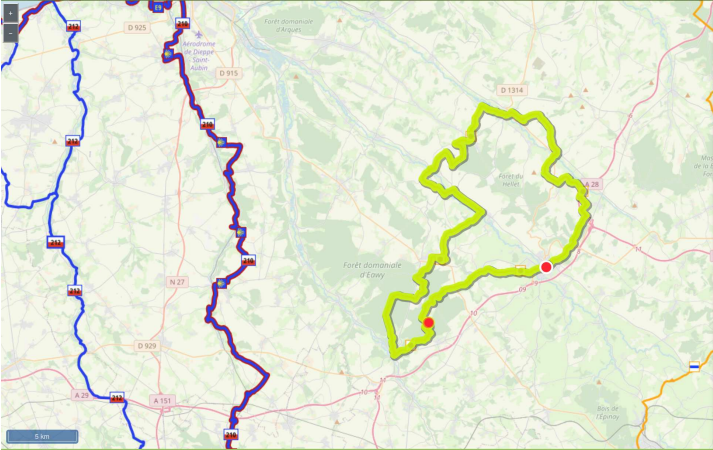
Routes

Analyse your relation

Waymarked Trails: Hiking

Last update: 11/2/25 2:45 PM

Map data © OpenStreetMap under ODbL. Base map: OpenStreetMap



Relation 8673891

► Elevation profile

▼ OpenStreetMap tags

key	value
colour	purple
name	Au Coeur du Talou
network	rwn
operator	SMAD du Pays de Bray
osmc:symbol	purple:white:purple_dot:black
roundtrip	yes
route	hiking
type	route

► Download route

▼ Analyze relation

This relation has 2 segments.

- Open end in the route.
- More than two segments of the route meet in this point.
- Point where route is interrupted because relation is not sorted. The red dashed lines shows where the route continues.

This shows the route segments as they were computed by waymarkedtrails from the relation, which may not always be correct. For more information how waymarkedtrails sorts routes, visit the help pages.

Show on map

Search

Settings Location Help

Routes

Analyse your relation

Waymarked Trails: Hiking

Last update: 11/2/25 2:45 PM Map data © OpenStreetMap under ODbL. Base map: OpenStreetMap

Relation 8673891

► Elevation profile

▼ OpenStreetMap tags

key	value
colour	purple
name	Au Coeur du Talou
network	rwn
operator	SMAD du Pays de Bray
osmc:symbol	purple:white:purple_dot:black
roundtrip	yes
route	hiking
type	route

► Download route

▼ Analyze relation

This relation has 2 segments.

- Open end in the route.
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- Point where route is interrupted because relation is not sorted. The red dashed lines shows where the route continues.

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Show on map

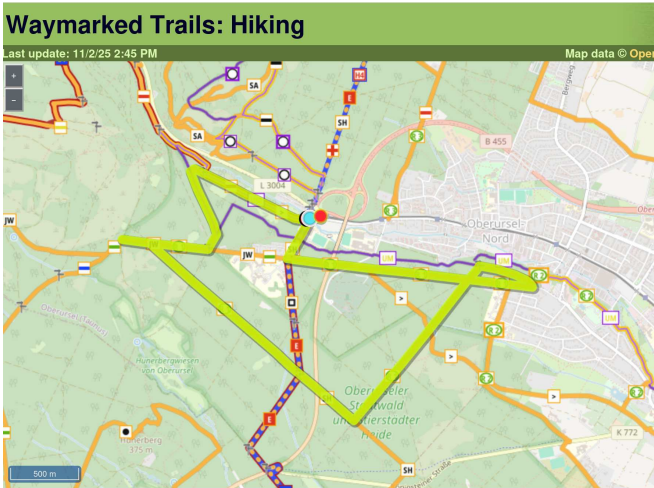
Routes

Analyse your relation

Waymarked Trails: Hiking

Last update: 11/2/25 2:45 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap



Relation 6390768

osmc:symbol	:green_circle::R 1:green
ref	R 1
roundtrip	yes
route	hiking
symbol	green R 1 on white ground
type	route

► Download route

▼ Analyze relation

This relation has 2 segments.

- Open end in the route.
- More than two segments of the route meet in this point.
- Point where route is interrupted because relation is not sorted. The red dashed lines shows where the route continues.

This shows the route segments as they were computed by waymarkedtrails from the relation, which may not always be correct. For more information how waymarkedtrails sorts routes, visit the help pages.

Show on map

Search

Routes

Analyse your relation

Waymarked Trails: Hiking

Last update: 11/2/25 2:45 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap



Relation 6390768

osmc:symbol	:green_circle::R 1:green
ref	R 1
roundtrip	yes
route	hiking
symbol	green R 1 on white ground
type	route

Download route

Analyze relation

This relation has 2 segments.

- Open end in the route.
- More than two segments of the route meet in this point.
- Point where route is interrupted because relation is not sorted. The red dashed lines shows where the route continues.

This shows the route segments as they were computed by waymarkedtrails from the relation, which may not always be correct. For more information how waymarkedtrails sorts routes, visit the help pages.

Show on map

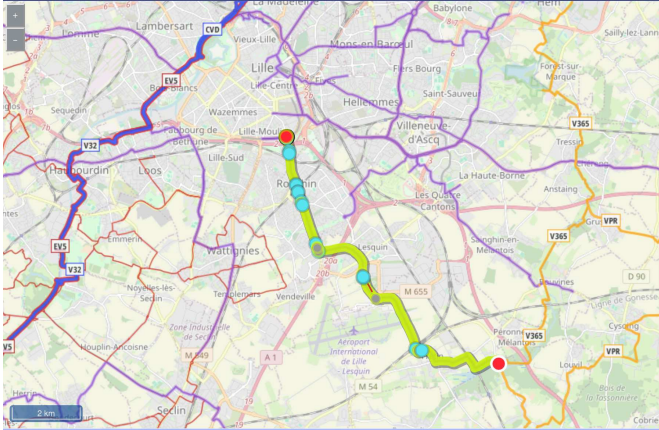
Routes

Analyse your relation

Waymarked Trails: Cycling

Last update: 11/2/25 2:54 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap



Relation 16291136

Liaison Lille <=> Fretin

LLF

Mapped length: 12 km

Operator: MEL

► Elevation profile

▼ OpenStreetMap tags

key	value
from	Lille
name	Liaison Lille <=> Fretin
network	lcn
operator	MEL
route	bicycle
to	Fretin
type	route

► Download route

▼ Analyze relation

Show on map

Search



Routes

Analyse your relation

Waymarked Trails: Cycling

Last update: 11/2/25 2:54 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap

Relation 16291136 ×

Liaison Lille <=> Fretin **LLF**

Mapped length: 12 km
Operator: MEL

► Elevation profile

▼ OpenStreetMap tags

key	value
from	Lille
name	Liaison Lille <=> Fretin
network	lcn
operator	MEL
route	bicycle
to	Fretin
type	route

► Download route

▼ Analyze relation

[Show on map](#)

Search

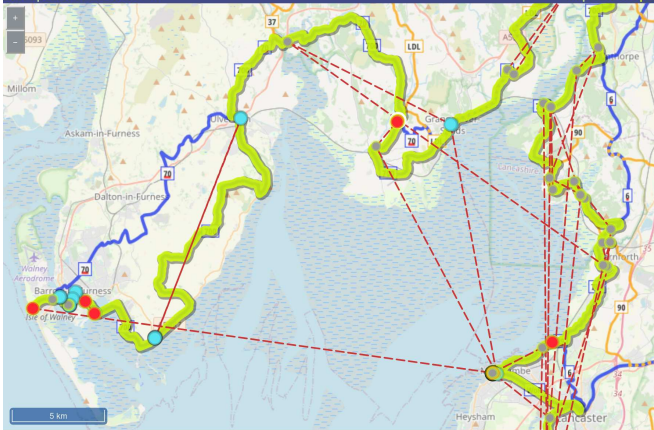
Routes

Analyse your relation

Waymarked Trails: Cycling

Last update: 11/2/25 3:08 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap



Relation 7024538

ref	700
route	bicycle
to	Walney Island
type	route
website	https://www.morecambebay.org.uk/be-inspired/cycling/bay-cycle-way

► Download route

▼ Analyze relation

This relation has 59 segments.

- Open end in the route.
- More than two segments of the route meet in this point.
- Point where route is interrupted because relation is not sorted. The red dashed lines shows where the route continues.

This shows the route segments as they were computed by waymarkedtrails from the relation, which may not always be correct. For more information how waymarkedtrails sorts routes, visit the help pages.

Show on map

Search

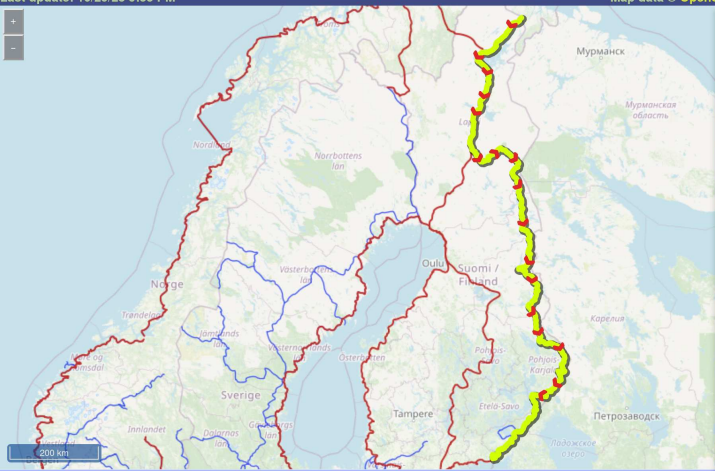
Routes

Why Bother with Route Roles: GPX Export

Waymarked Trails: Cycling

Last update: 10/29/25 9:55 PM

Map data © OpenStreetMap under ODbL Base map: OpenStreetMap



Relation 15835151 ×

[Website](#)

► **Elevation profile**

► **Part of**

► **OpenStreetMap tags**

▼ **Download route**

Format

☒ GPX ☐ KML

Include ways in direction

☒ forwards
☐ backwards
☐ both

Additional route parts

☐ alternative ☐ connection

Get file

► **Analyze relation**

[Show on map](#)

Thank you.



<https://waymarkedtrails.org>